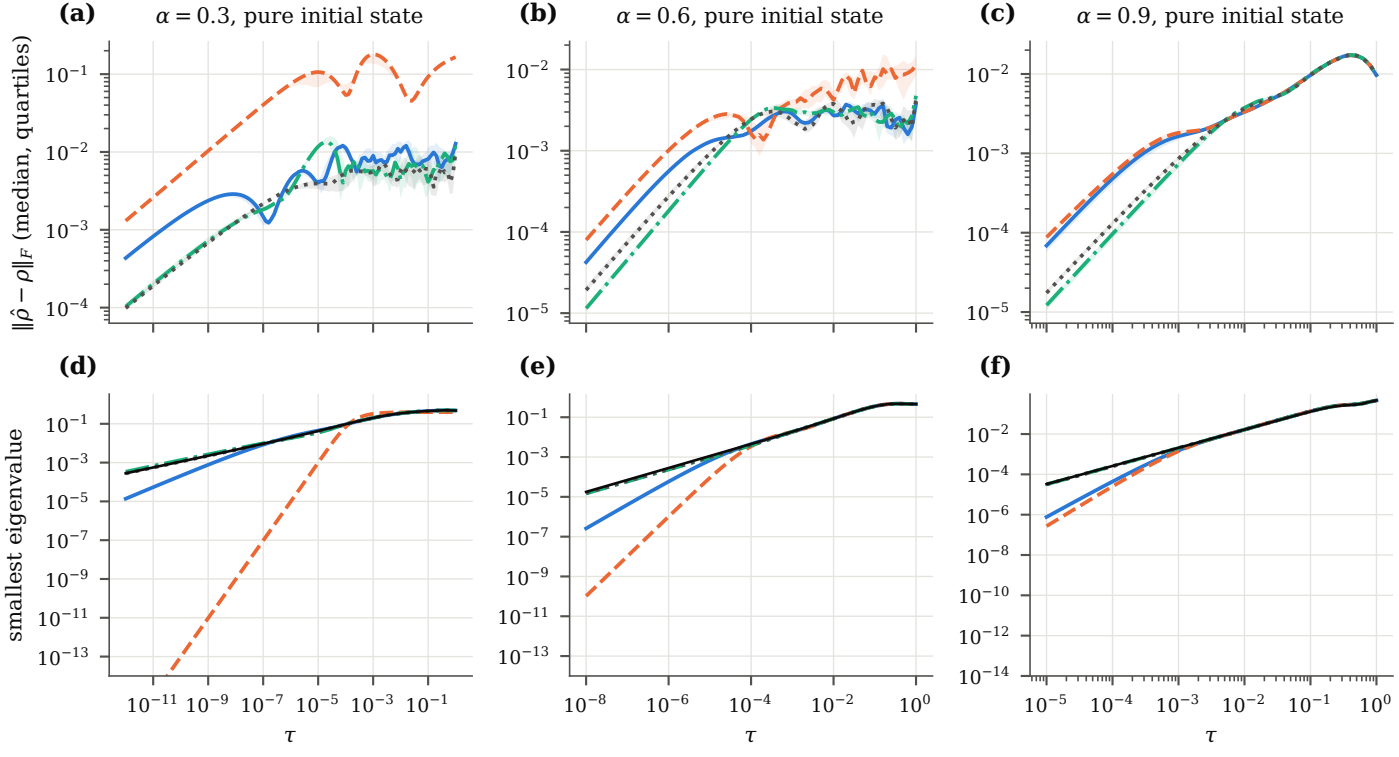

(a)
$\alpha = 0.3$, pure initial state
(b)
$\alpha = 0.6$, pure initial state
(c)
$\alpha = 0.9$, pure initial state
$\|\hat{\rho} - \rho\|_F$ (median, quartiles)
(d)
(e)
(f)
smallest eigenvalue
τ
τ
τ
PSD factor, τ^α gate
PSD factor, gate $1 - e^{-\beta\tau}$
linear head, τ^α gate
PSD block gate, τ^α gate
exact